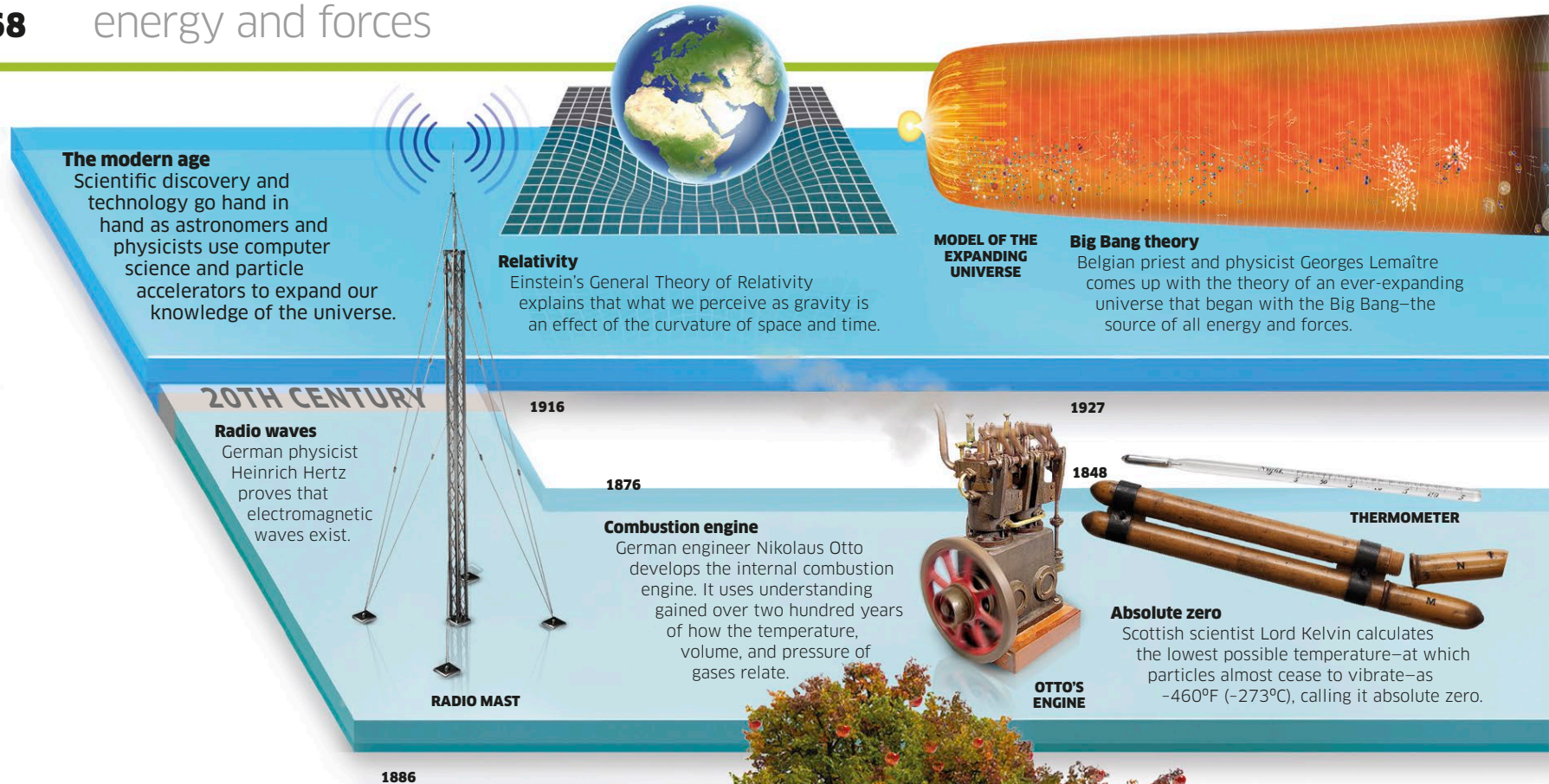




Discovering energy and forces

People have been asking questions about how the world around them works, and using science to find answers for them, for thousands of years.

From the forces that keep a ship afloat and the magnetism that helps sailors to navigate the oceans with a compass, to the atoms and subatomic particles that make up our world and the vast expanses of space, people through history have learned about the universe by observation and experiment. In ancient and medieval times, as the tools available to study the world were limited, so was knowledge of science. The modern scientific method is based on experiments, which are used to test hypotheses (unproven ideas). Observed results modify hypotheses, improving our understanding of science.

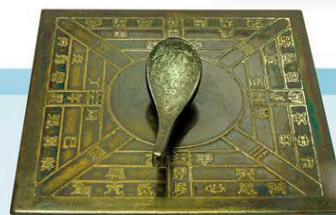
Ancient and medieval ideas

The ancient Greeks and Romans used debate to help them understand the universe, while Arab and Chinese scholars studied mathematics and natural phenomena such as rainbows and eclipses.



Buoyancy

The Greek thinker Archimedes realizes the force pushing upward on an object in water is equal to the weight of water displaced.



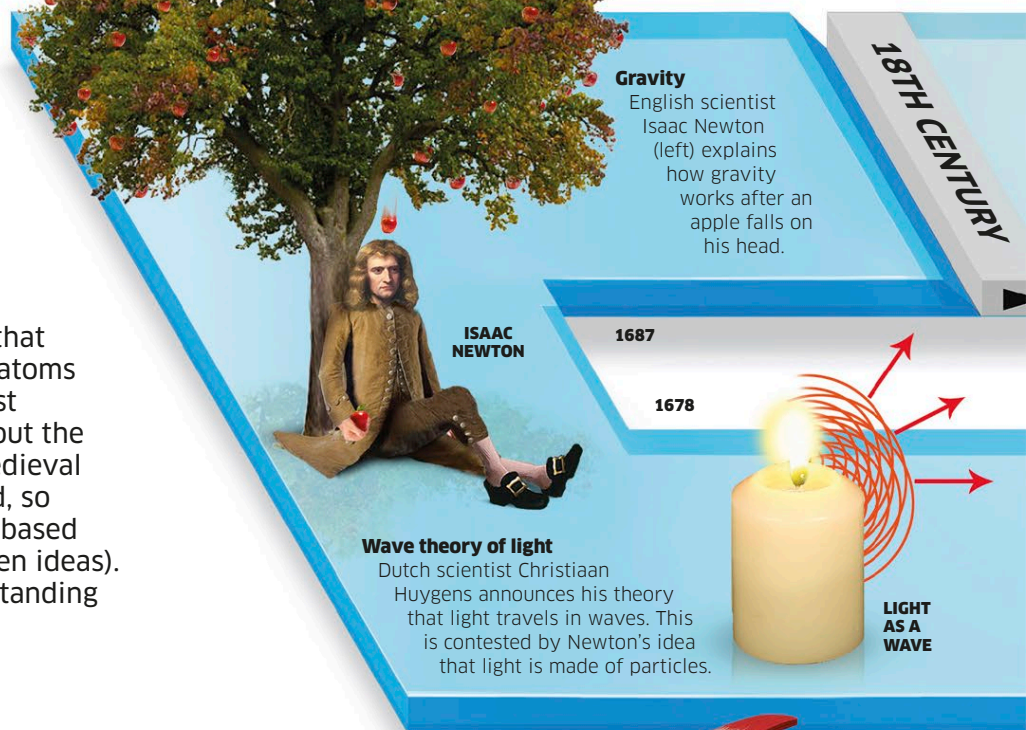
Magnetic compass

The Chinese create primitive compasses with lodestone, a naturally occurring magnet.



Light vision

Arab scholar Alhazen suggests that light is emitted from objects into the eye, not the reverse.



BEFORE 1500

240 BCE

200 BCE

1011 CE